

MCA II

SEM III

Data Ware Housing and Data Mining IT-32



Data Warehouse and Data Mining

Chapter 2

Data Warehouse Architecture

Dr. Ashwini Brahme

Data Warehouse Architecture

Three-tier architecture: This is the most widely used architecture. It consists of the,

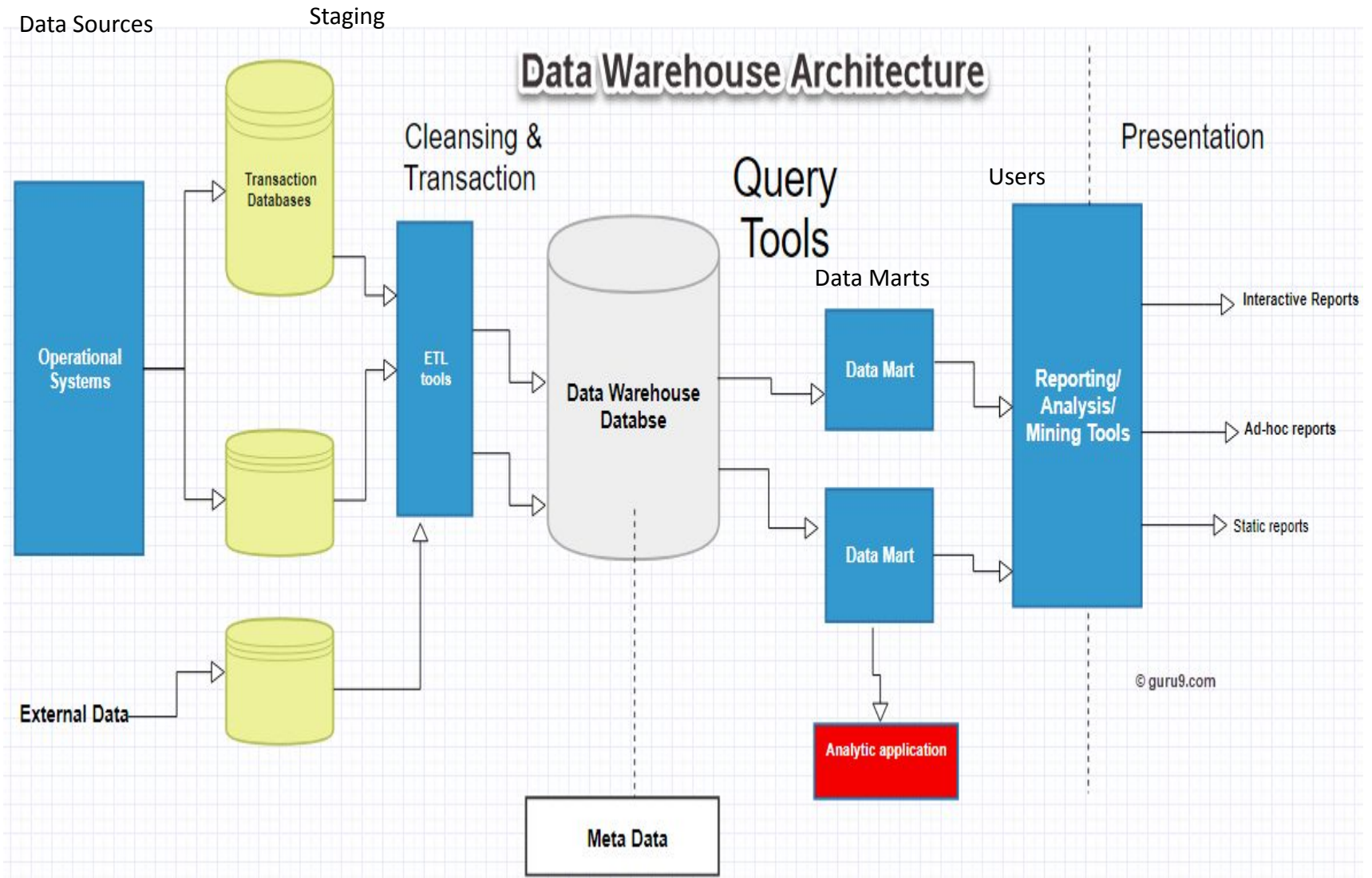
1. Top,
2. Middle and
3. Bottom Tier.

Bottom Tier: The database of the Data warehouse servers as the bottom tier. It is usually a relational database system. Data is cleansed, transformed, and loaded into this layer using back-end tools.

Middle Tier: The middle tier in Data warehouse is **an OLAP server** which is implemented using either ROLAP or MOLAP model. For a user, this application tier presents an abstracted view of the database. This layer also acts as a mediator between the end-user and the database.

Top-Tier: The top tier is a **front-end client layer**. Top tier is the tools and API that you connect and get data out from the data warehouse. It could be Query tools, reporting tools, managed query tools, Analysis tools and Data mining tools.

Tree-Tier Architecture



Data Warehouse Architecture

There are mainly Nine components of Data Warehouse

1. Data Base/Data Source
 2. Data Extraction
 3. Staging
 4. Sourcing, Acquisition, Clean-up and Transformation Tools (ETL)
 5. Data Storage
 6. Data Logic
 7. Data Presentation
 8. Meta Data
 9. Query Tool/System Operation
- The process of gathering, cleaning and integrating data from various sources, usually from long-term existing operational systems is in part for each environment. The operational systems are frequently reexamined as new decision support a requirement emerges. Often new requirements require gathering, cleaning and integrating new data from "data marts" which is tailored for instant access by users.
 - Data warehouse which is also known as an **enterprise data warehouse** is a system used for reporting and data analysis. It is also considered as a core component of business intelligence. The data stored in the warehouse is uploaded from the operational systems. The data passes through an operational data store and will require data cleansing for additional operations to ensure data quality before it is used in the Data Warehouse for reporting.

Data Warehouse Architecture

- The typical **Extract, transform, load (ETL)**- based data warehouse uses staging, data integration, and access layers to express its key functions.
- The staging layer database stores raw data taken from each of the disparate source data systems.
- The integration layer integrates the disparate data sets by transforming the data from the staging layer often storing this transformed data in an operational data store (ODS) database.
- The integrated data are then moved to yet another database, often called the data warehouse database, where the data is arranged into hierarchical groups and into facts and aggregate facts.
- Star schema is the name coined for the combination of facts and dimensions the users can retrieve data through access layer

Data Warehouse Architecture

- The main source of the data is cleansed, transformed, catalogued and made available for use by managers and other business professionals for data mining, online analytical processing, market research and decision support.
- The method to retrieve and analyze data, to extract, transforms, and load data, and to manage the data dictionary are also important for a data warehousing system.
- An expanded definition for data warehousing includes business intelligence tools, tools to extract, transform and load data into the repository and tools to manage and retrieve metadata.
- Data warehousing emphasizes the capture of data from data mart diverse sources for useful analysis and access start from the point-of-view of the end user who may need access to specialized, sometimes local databases.

General stages of Data Warehouse

- General Stages of Data Warehouse are as follows,
 1. Offline Operational Database
 2. Offline Data Warehouse
 3. Real Time Data Warehouse
 4. Integrated Data Warehouse
- In **Offline Operational Database** the data is transformed from operational system to a server. Processing, reporting and loading of the copied data over impact the performance of the operational systems
- In **Offline Data Warehouse** data is frequently updated from the Operational Database. The transformation of data in the Data warehouse is done to meet the objectives of Data warehouse
- During **Real time Data Warehouse** stage every data warehouses are refreshed and updated during a transaction. For instance this is done when a ticket is booked in an airline or in a railway booking system.
- In **Integrated Data Warehouse** stage leads to the next process that is when an transaction takes place. The transaction is generated through the Data Warehouses to the operational system

Components of Data warehouse

- **Components of Data warehouse are as follows**
 1. Load manager
 2. Data Warehouse Manager
 3. Query Manager
 4. End-user Access Tools
- **Load Manager** extract an input data into the warehouse its otherwise called as the front component. It also transforms the preparation of data to enter in to a data.
- **Data Warehouse Manager** does operation which is closely associated with managing the data in the warehouse. It analysis data to conform creation of induces and view consistency, de-normalization and aggregations
- **Query Manager** is very well known as backend component. It assists the entire operation operations related to the user queries.
- **End-user access tools** includes five different groups like
 1. Data Reporting
 2. Query Tools
 3. Application development tools
 4. EIS tools,
 5. OLAP tools and data mining tools.

Federated Data warehouse Architecture

- A federated data warehouse is a practical approach to achieving the “single version of the truth” across the organization. The federated data warehouse is used to integrate key business measures and dimensions. The foundations of the federated data warehouse are the common business model and common staging area.
- Two approaches :
- Regional federation
- Functional federation

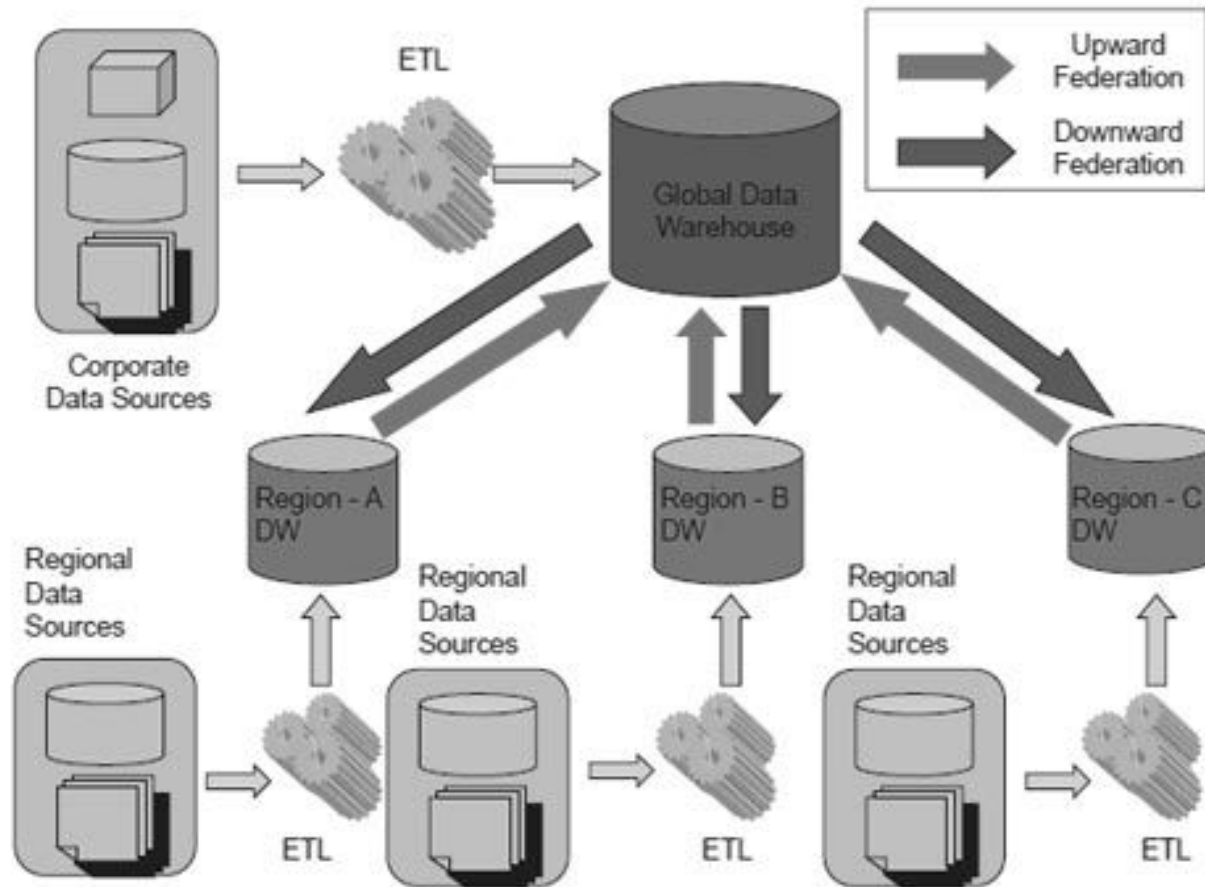
Regional federation

- The big organization has various regions that provide businesses to customers globally.
- Different regional data warehouses were built for each region to meet the specific business needs. A global data warehouse also was built to provide analytical capabilities to the executive at the global level.

Regional Federation – Federated Data Warehouse

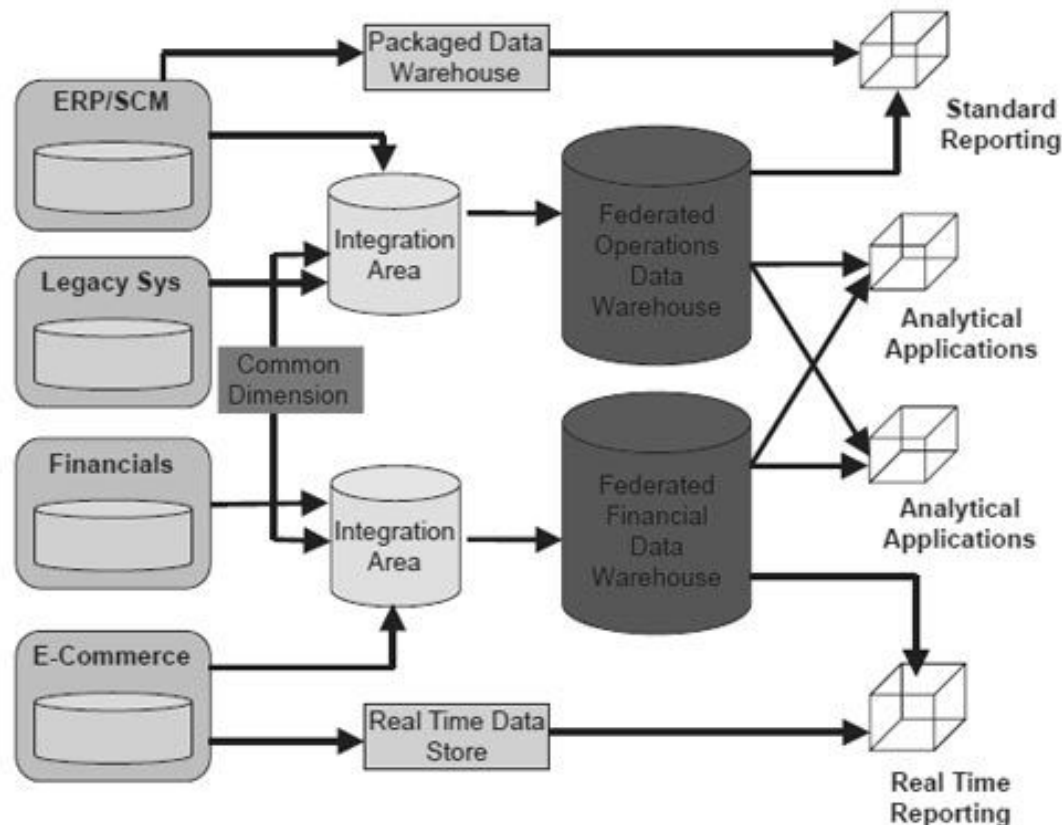
- The difference between the regional and global data warehouse systems is the nature of data resided at each system level.
- there are two data flows between regional and global data warehouses:
- **Upward federation** – **only fact data are moved from** regional data warehouse to global data warehouse.
- The **aggregation of data can take place at a global** data warehouse after data integrated or during data movement.
- **Downward federation** – in the downward federation, the reference flows **from the global to the regional level.**
- This ensures the consistency and integrity of data across the organization.
- Transactional data from corporate operational systems such as ERP and CRM are sourced at the global level and then extracted, transformed, and loaded into a respective regional data warehouse.

Regional Federation – Federated Data Warehouse



Functional Federation – Federated Data Warehouse

- A functional federated data warehouse is used **when the organizations have different data warehouses system** was built for specific applications such as ERP, CRM, or subject-specific.
- The components of functional federated data warehouse architecture include **data marts, custom-built data warehouses, ETL tools, cross-function reporting systems, real-time data store, and reporting** as the picture below:



Advantages of Federated Data warehouse Architecture

- **Ease of implementation** – Federated data warehouse integrated all legacy data warehouses, business intelligence systems into a newer system that provides analytical capabilities across the function. Federated data warehouse data do not try to rebuild a new system which potentially causes the major point of conflict.
- **Shorter implementation time** – By integrating all legacy BI systems, the federated data warehouse approach has a shorter implementation time in comparison with the lengthy processing of building an enterprise data warehouse.
- **Cross-functional analytics requirements** – Cross function analytics requirements accomplished using common business modules across different BI systems of each department. A Federated data warehouse is the dynamic cooperation of various business intelligence systems to make them talk to each other.
- A Federated data warehouse **offers a practical solution for building the data warehouse**. The iterative manner of the federated data warehouse approach helps **reduce the implementation time and cost, therefore, provide an excellent proposition to the business**.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

- **ER Modeling**

- **Data and Objects:**
- **Data:** Facts and statistic collected for reference or analysis.
- **Data objects:** Data objects represent an entity. They can also be referred to as samples, examples, instances, data points, or objects. They are stored in a database in form data tuples i.e. the rows of a database correspond to the data objects, and the columns correspond to the attributes. Following are some examples of objects.

In a sales database, the objects may be customers, store items, and sales;

In a medical database, the objects may be patients;

In a university database, the objects may be students, professors, and courses.

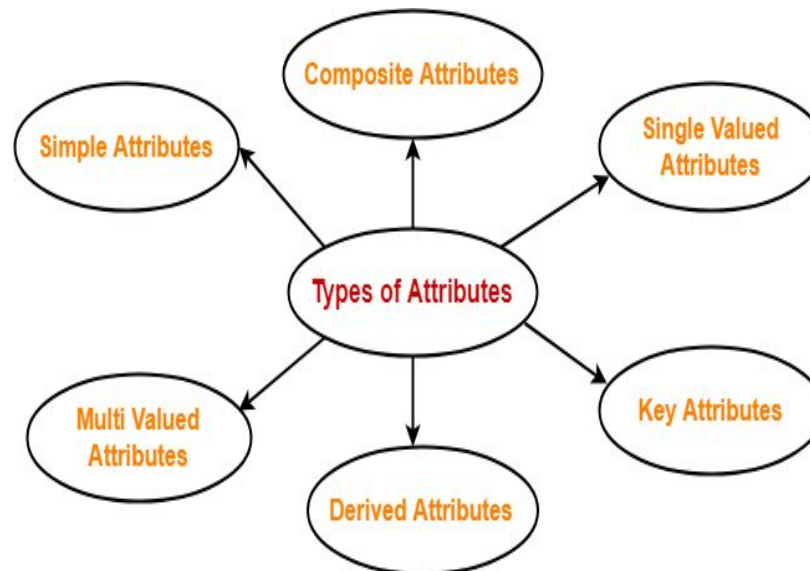
- **Attributes:** Data objects are typically described by attributes. Attributes are the descriptive properties which are owned by each entity of an Entity Set. There exist a specific domain or set of values for each attribute from where the attribute can take its values.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

- **ER Modeling**

- Types of Attributes: Attributes are of Following Types

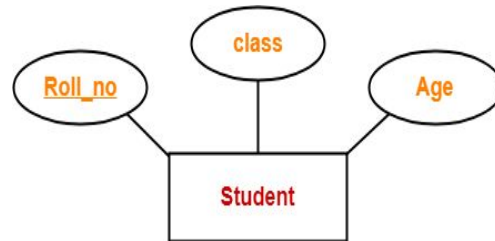
1. Simple attributes
2. Composite attributes
3. Single valued attributes
4. Multi valued attributes
5. Derived attributes
6. Key attributes



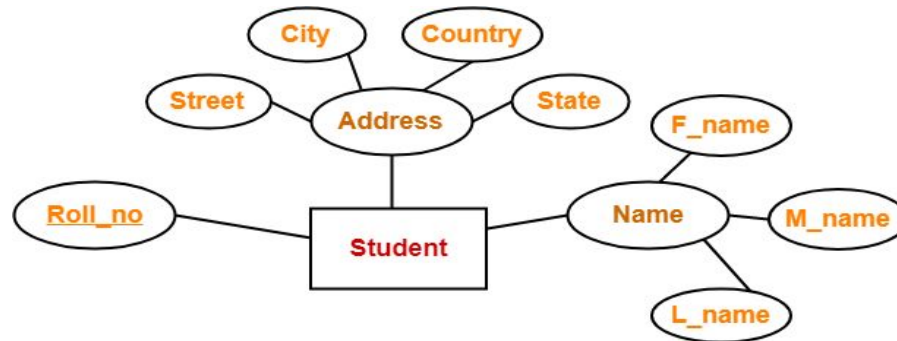
Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

- **ER Modeling**

1. Simple Attributes: Simple attributes are those attributes which cannot be divided further.



- **2. Composite Attributes-** Composite attributes are those attributes which are composed of many other simple attributes.

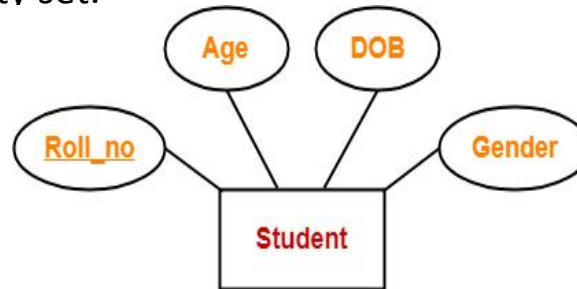


- Here, the attributes “Name” and “Address” are composite attributes as they are composed of many other simple attributes.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

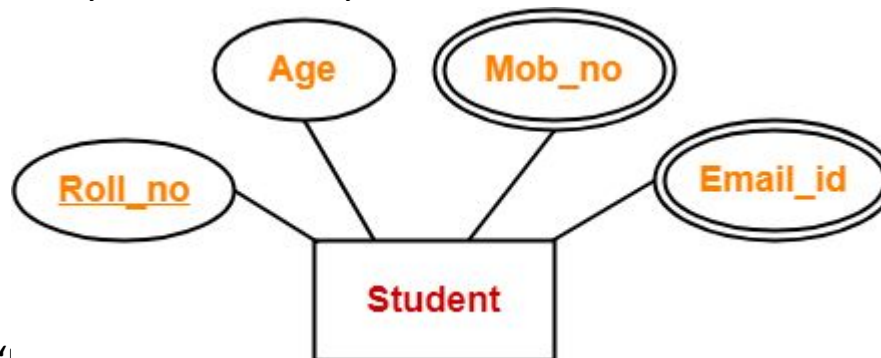
ER Modeling

3. **Single Valued Attributes**-Single valued attributes are those attributes which can take only one value for a given entity from an entity set.



Here, all the attributes are single valued attributes as they can take only one specific value for each entity.

4. **Multi Valued Attributes**- Multi valued attributes are those attributes which can take more than one value for a given entity from an entity set.

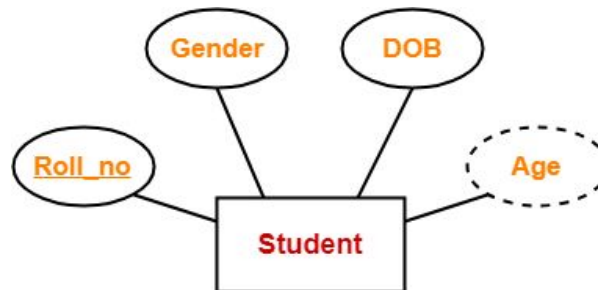


• Here, the attributes "Mob_no" and "Email_id" are multi valued attributes as they can take more than one values for a given entity.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

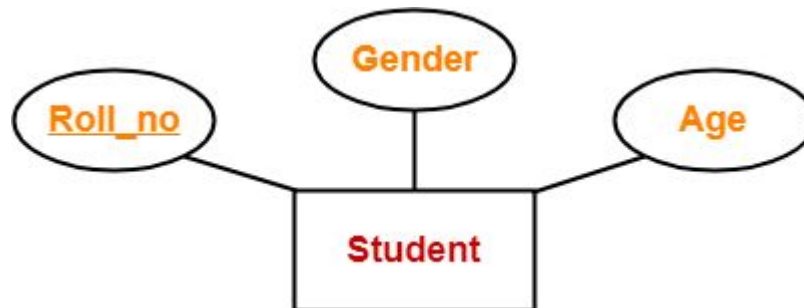
ER Modeling

5. Derived Attributes-Derived attributes are those attributes which can be derived from other attribute(s).



Here, the attribute “Age” is a derived attribute as it can be derived from the attribute “DOB”.

6. Key Attributes- Key attributes are those attributes which can identify an entity uniquely in an entity set.



Here, the attribute “Roll_no” is a key attribute as it can identify any student uniquely.

Data Warehouse Schemas

- Star Schema
- Inside Dimension Table
- Inside Fact Table
- Fact less Fact Table
- Granularity
- Star Schema Keys
- Snowflake Schema
- Fact Consolation Schema

Introduction to Meta Data

- Federated Data

Meta Data Management in practice

- Metadata requirement gathering
- Metadata Classification
- Meta data collection strategies
- Tools for Metadata management

Steps to Implement Data Warehouse

- Data Warehouse implementation is to employ a three-prong strategy as below
 1. Enterprise Strategy
 2. Phased Delivery
 3. Iterative Prototyping
- **Enterprise strategy:** Data transformation and data mapping is fast through this strategy, and at the same time attributes, dimension and fact are identified
- **Phased delivery:** Data warehouse implementation is phased based upon subject areas. For instance business steps like booking and billing are first implemented and then integrated with one another.
- **Iterative Prototyping:** In this method data warehouse is developed and tested iteratively

Best practices to implement a Data Warehouse

- First decide a plan to accurately identify the consistency and integrity of the data.
- The data warehouse should be, well defined and well integrated
- While designing Data warehouse make sure you use right tool, stick to life cycle, take care about data conflicts and ready to learn from your mistakes.
- Don't replace reports and operational systems
- When designing data warehouse, conform that the right tool is being used and don't spend much time on cleaning extracting and loading data.
- Identify and implement a training plan for the end users
- Try to involve all stakeholder and business partners in data warehouse implantation process.

Future of Data Warehousing

- The scope of data warehousing are on the rise as the regulatory constraints changes from time to time.
- It may limit the performance to combine different data from different sources. This may include unstructured data which is usually a difficult one to store.
- This is because the size of the data grows everyday data warehouse system helps the user to build and run data as a size grows

Conclusion

- A data warehouse is a database, which is kept separate from the organization's operational database.
- There is no frequent updating done in a data warehouse.
- It possesses consolidated historical data, which helps the organization to analyze its business.
- A data warehouse helps executives to organize, understand, and use their data to take strategic decisions.
- Data warehouse systems help in the integration of diversity of application systems.
- A data warehouse system helps in consolidated historical data analysis.
- Data mining functions such as association, clustering, classification, prediction can be integrated with OLAP operations to enhance the interactive mining of knowledge at multiple level of abstraction.

Thank You

MCA II

SEM III

Data Ware Housing and Data Mining IT-32



Data Warehouse and Data Mining

Chapter 2

Data Warehouse Architecture

Dr. Ashwini Brahme

Data Warehouse Architecture

Three-tier architecture: This is the most widely used architecture. It consists of the,

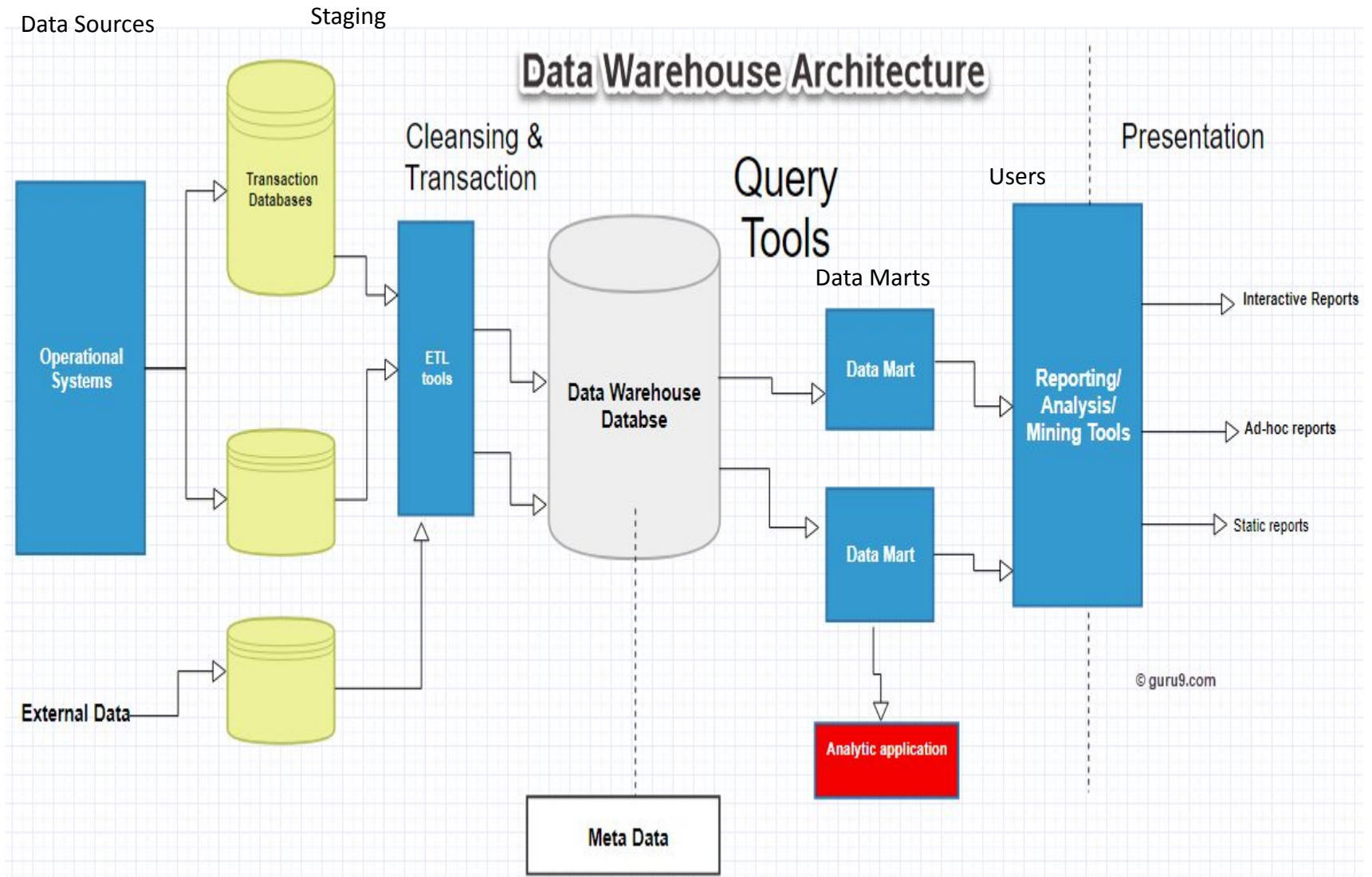
1. Top,
2. Middle and
3. Bottom Tier.

Bottom Tier: The database of the Data warehouse servers as the bottom tier. It is usually a relational database system. Data is cleansed, transformed, and loaded into this layer using back-end tools.

Middle Tier: The middle tier in Data warehouse is **an OLAP server** which is implemented using either ROLAP or MOLAP model. For a user, this application tier presents an abstracted view of the database. This layer also acts as a mediator between the end-user and the database.

Top-Tier: The top tier is a **front-end client layer**. Top tier is the tools and API that you connect and get data out from the data warehouse. It could be Query tools, reporting tools, managed query tools, Analysis tools and Data mining tools.

Tree-Tier Architecture



Data Warehouse Architecture

There are mainly Nine components of Data Warehouse

1. Data Base/Data Source
 2. Data Extraction
 3. Staging
 4. Sourcing, Acquisition, Clean-up and Transformation Tools (ETL)
 5. Data Storage
 6. Data Logic
 7. Data Presentation
 8. Meta Data
 9. Query Tool/System Operation
- The process of gathering, cleaning and integrating data from various sources, usually from long-term existing operational systems is in part for each environment. The operational systems are frequently reexamined as new decision support a requirement emerges. Often new requirements require gathering, cleaning and integrating new data from "data marts" which is tailored for instant access by users.
 - Data warehouse which is also known as an **enterprise data warehouse** is a system used for reporting and data analysis. It is also considered as a core component of business intelligence. The data stored in the warehouse is uploaded from the operational systems. The data passes through an operational data store and will require data cleansing for additional operations to ensure data quality before it is used in the Data Warehouse for reporting.

Data Warehouse Architecture

- The typical **Extract, transform, load (ETL)**- based data warehouse uses staging, data integration, and access layers to express its key functions.
- The staging layer database stores raw data taken from each of the disparate source data systems.
- The integration layer integrates the disparate data sets by transforming the data from the staging layer often storing this transformed data in an operational data store (ODS) database.
- The integrated data are then moved to yet another database, often called the data warehouse database, where the data is arranged into hierarchical groups and into facts and aggregate facts.
- Star schema is the name coined for the combination of facts and dimensions the users can retrieve data through access layer

Data Warehouse Architecture

- The main source of the data is cleansed, transformed, catalogued and made available for use by managers and other business professionals for data mining, online analytical processing, market research and decision support.
- The method to retrieve and analyze data, to extract, transforms, and load data, and to manage the data dictionary are also important for a data warehousing system.
- An expanded definition for data warehousing includes business intelligence tools, tools to extract, transform and load data into the repository and tools to manage and retrieve metadata.
- Data warehousing emphasizes the capture of data from data mart diverse sources for useful analysis and access start from the point-of-view of the end user who may need access to specialized, sometimes local databases.

General stages of Data Warehouse

- General Stages of Data Warehouse are as follows,
 1. Offline Operational Database
 2. Offline Data Warehouse
 3. Real Time Data Warehouse
 4. Integrated Data Warehouse
- In **Offline Operational Database** the data is transformed from operational system to a server. Processing, reporting and loading of the copied data over impact the performance of the operational systems
- In **Offline Data Warehouse** data is frequently updated from the Operational Database. The transformation of data in the Data warehouse is done to meet the objectives of Data warehouse
- During **Real time Data Warehouse** stage every data warehouses are refreshed and updated during a transaction. For instance this is done when a ticket is booked in an airline or in a railway booking system.
- In **Integrated Data Warehouse** stage leads to the next process that is when an transaction takes place. The transaction is generated through the Data Warehouses to the operational system

Components of Data warehouse

- **Components of Data warehouse are as follows**
 1. Load manager
 2. Data Warehouse Manager
 3. Query Manager
 4. End-user Access Tools
- **Load Manager** extract an input data into the warehouse its otherwise called as the front component. It also transforms the preparation of data to enter in to a data.
- **Data Warehouse Manager** does operation which is closely associated with managing the data in the warehouse. It analysis data to conform creation of induces and view consistency, de-normalization and aggregations
- **Query Manager** is very well known as backend component. It assists the entire operation operations related to the user queries.
- **End-user access tools** includes five different groups like
 1. Data Reporting
 2. Query Tools
 3. Application development tools
 4. EIS tools,
 5. OLAP tools and data mining tools.

Federated Data warehouse Architecture

- A federated data warehouse is a practical approach to achieving the “single version of the truth” across the organization. The federated data warehouse is used to integrate key business measures and dimensions. The foundations of the federated data warehouse are the common business model and common staging area.
- Two approaches :
- Regional federation
- Functional federation

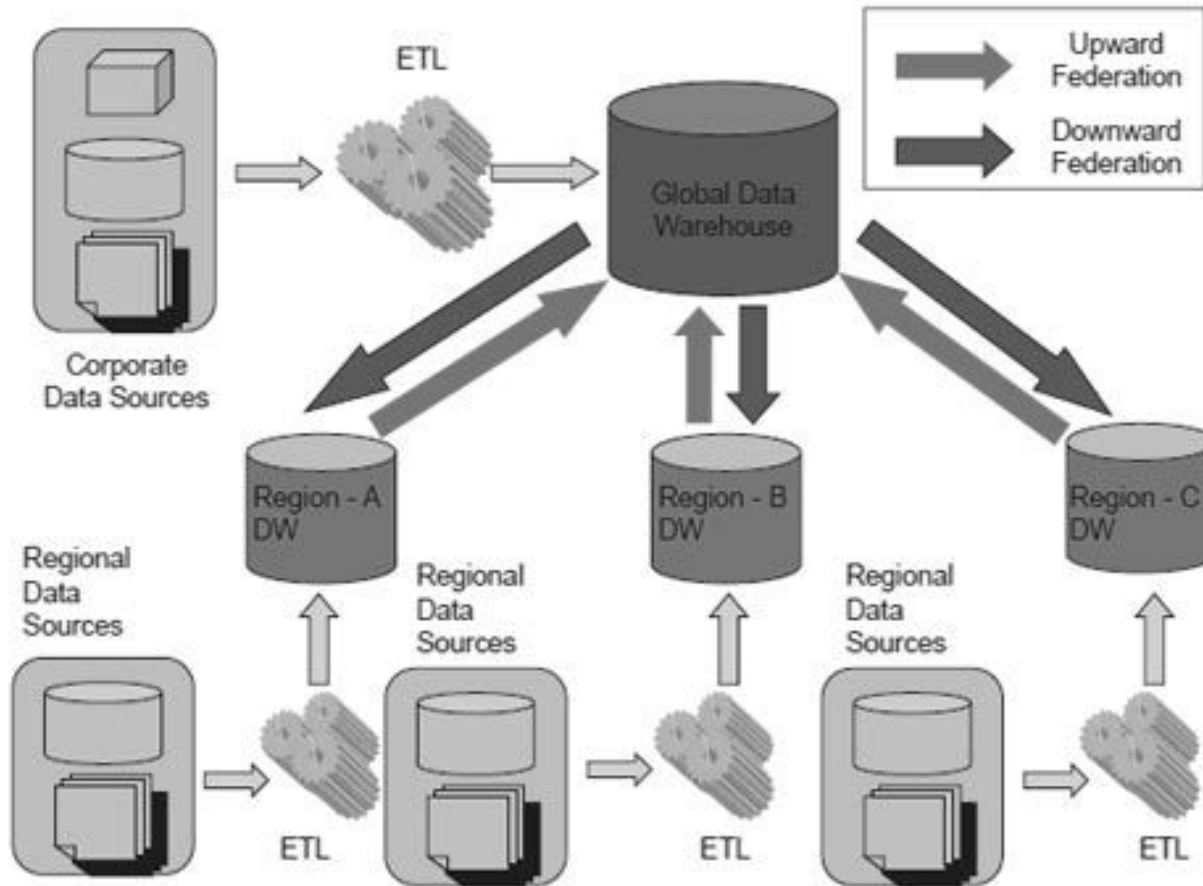
Regional federation

- The big organization has various regions that provide businesses to customers globally.
- Different regional data warehouses were built for each region to meet the specific business needs. A global data warehouse also was built to provide analytical capabilities to the executive at the global level.

Regional Federation – Federated Data Warehouse

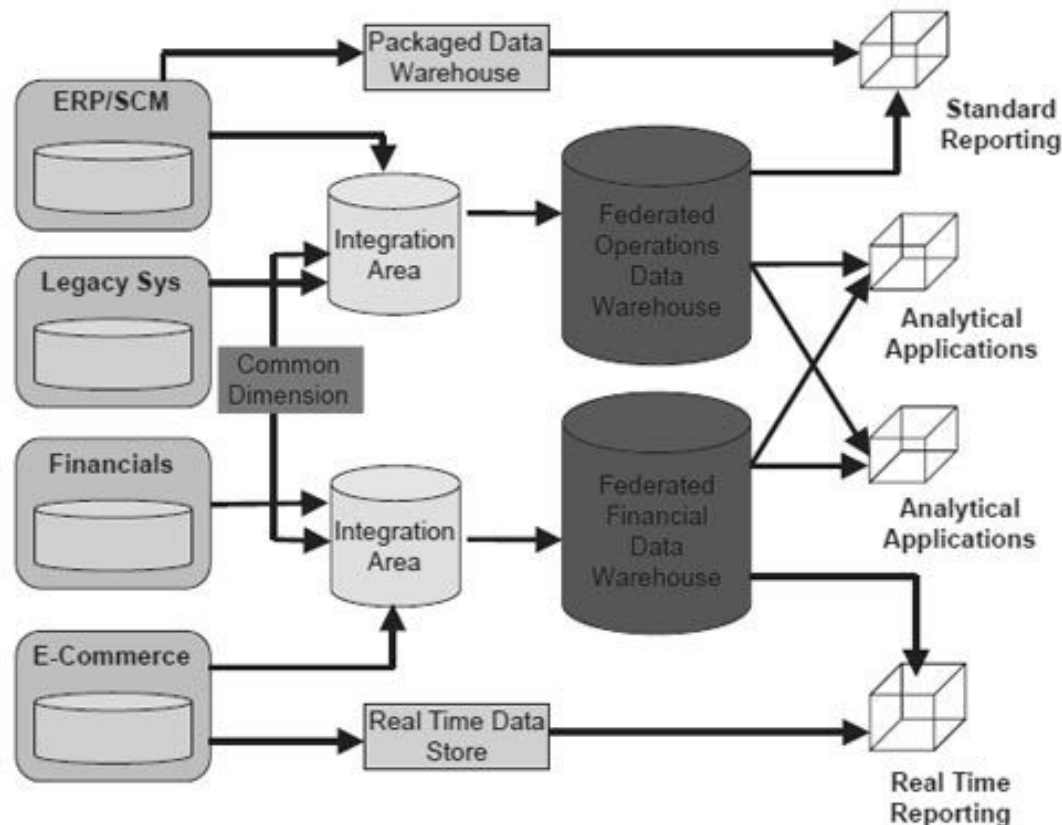
- The difference between the regional and global data warehouse systems is the nature of data resided at each system level.
- there are two data flows between regional and global data warehouses:
- **Upward federation** – **only fact data are moved from** regional data warehouse to global data warehouse.
- The **aggregation of data can take place at a global** data warehouse after data integrated or during data movement.
- **Downward federation** – in the downward federation, the reference flows **from the global to the regional level.**
- This ensures the consistency and integrity of data across the organization.
- Transactional data from corporate operational systems such as ERP and CRM are sourced at the global level and then extracted, transformed, and loaded into a respective regional data warehouse.

Regional Federation – Federated Data Warehouse



Functional Federation – Federated Data Warehouse

- A functional federated data warehouse is used **when the organizations have different data warehouses system** was built for specific applications such as ERP, CRM, or subject-specific.
- The components of functional federated data warehouse architecture include **data marts, custom-built data warehouses, ETL tools, cross-function reporting systems, real-time data store, and reporting** as the picture below:



Advantages of Federated Data warehouse Architecture

- **Ease of implementation** – Federated data warehouse integrated all legacy data warehouses, business intelligence systems into a newer system that provides analytical capabilities across the function. Federated data warehouse data do not try to rebuild a new system which potentially causes the major point of conflict.
- **Shorter implementation time** – By integrating all legacy BI systems, the federated data warehouse approach has a shorter implementation time in comparison with the lengthy processing of building an enterprise data warehouse.
- **Cross-functional analytics requirements** – Cross function analytics requirements accomplished using common business modules across different BI systems of each department. A Federated data warehouse is the dynamic cooperation of various business intelligence systems to make them talk to each other.
- A Federated data warehouse **offers a practical solution for building the data warehouse**. The iterative manner of the federated data warehouse approach helps **reduce the implementation time and cost, therefore, provide an excellent proposition to the business**.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

- **ER Modeling**

- **Data and Objects:**
- **Data:** Facts and statistic collected for reference or analysis.
- **Data objects:** Data objects represent an entity. They can also be referred to as samples, examples, instances, data points, or objects. They are stored in a database in form data tuples i.e. the rows of a database correspond to the data objects, and the columns correspond to the attributes. Following are some examples of objects.

In a sales database, the objects may be customers, store items, and sales;

In a medical database, the objects may be patients;

In a university database, the objects may be students, professors, and courses.

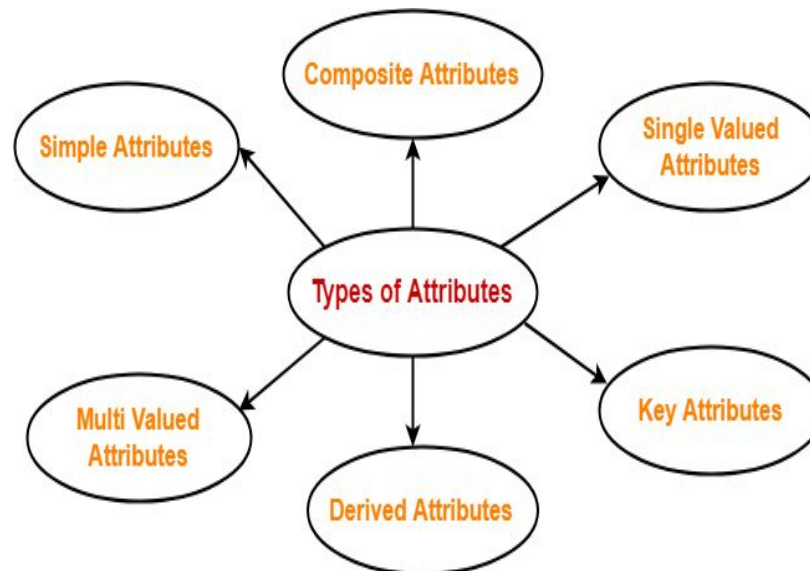
- **Attributes:** Data objects are typically described by attributes. Attributes are the descriptive properties which are owned by each entity of an Entity Set. There exist a specific domain or set of values for each attribute from where the attribute can take its values.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

- **ER Modeling**

- Types of Attributes: Attributes are of Following Types

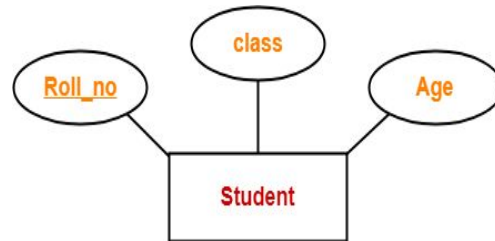
1. Simple attributes
2. Composite attributes
3. Single valued attributes
4. Multi valued attributes
5. Derived attributes
6. Key attributes



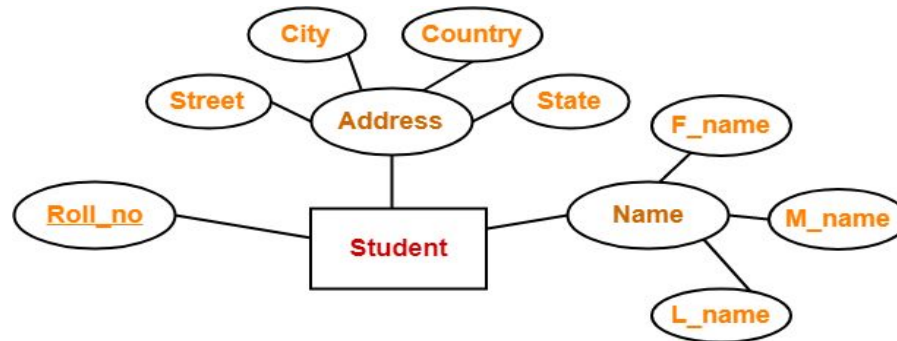
Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

- **ER Modeling**

1. Simple Attributes: Simple attributes are those attributes which cannot be divided further.



- **2. Composite Attributes-** Composite attributes are those attributes which are composed of many other simple attributes.

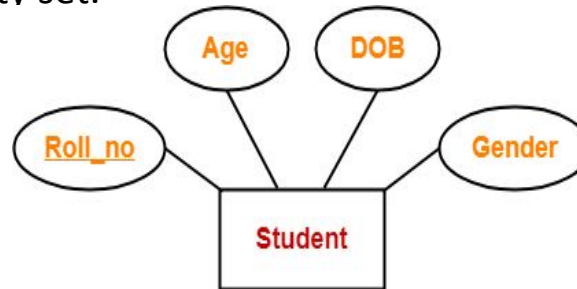


- Here, the attributes “Name” and “Address” are composite attributes as they are composed of many other simple attributes.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

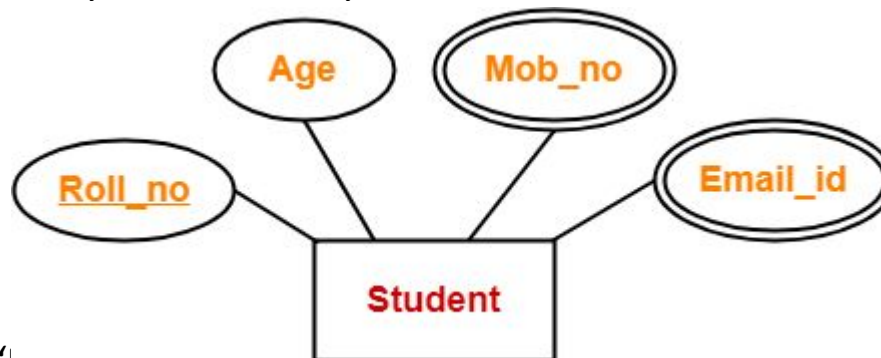
ER Modeling

3. **Single Valued Attributes**-Single valued attributes are those attributes which can take only one value for a given entity from an entity set.



Here, all the attributes are single valued attributes as they can take only one specific value for each entity.

4. **Multi Valued Attributes**- Multi valued attributes are those attributes which can take more than one value for a given entity from an entity set.

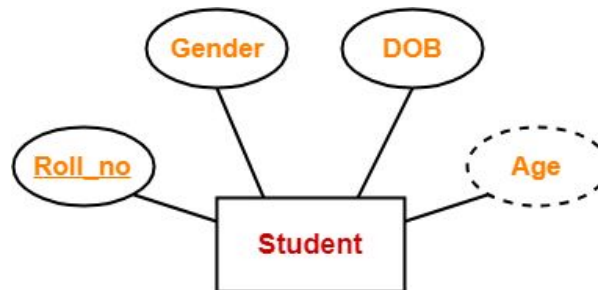


• Here, the attributes "Mob_no" and "Email_id" are multi valued attributes as they can take more than one values for a given entity.

Dimensional Modeling : ER Modeling Vs. Dimensional Modeling

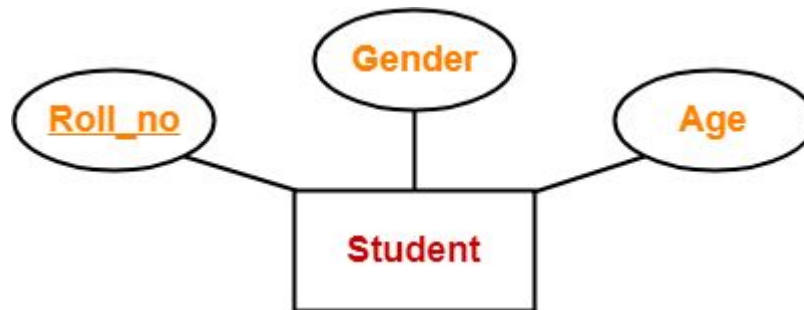
ER Modeling

5. Derived Attributes-Derived attributes are those attributes which can be derived from other attribute(s).



Here, the attribute “Age” is a derived attribute as it can be derived from the attribute “DOB”.

6. Key Attributes- Key attributes are those attributes which can identify an entity uniquely in an entity set.



Here, the attribute “Roll_no” is a key attribute as it can identify any student uniquely.

Data Warehouse Schemas

- Star Schema
- Inside Dimension Table
- Inside Fact Table
- Fact less Fact Table
- Granularity
- Star Schema Keys
- Snowflake Schema
- Fact Consolation Schema

Introduction to Meta Data

- Federated Data

Meta Data Management in practice

- Metadata requirement gathering
- Metadata Classification
- Meta data collection strategies
- Tools for Metadata management

Steps to Implement Data Warehouse

- Data Warehouse implementation is to employ a three-prong strategy as below
 1. Enterprise Strategy
 2. Phased Delivery
 3. Iterative Prototyping
- **Enterprise strategy:** Data transformation and data mapping is fast through this strategy, and at the same time attributes, dimension and fact are identified
- **Phased delivery:** Data warehouse implementation is phased based upon subject areas. For instance business steps like booking and billing are first implemented and then integrated with one another.
- **Iterative Prototyping:** In this method data warehouse is developed and tested iteratively

Best practices to implement a Data Warehouse

- First decide a plan to accurately identify the consistency and integrity of the data.
- The data warehouse should be, well defined and well integrated
- While designing Data warehouse make sure you use right tool, stick to life cycle, take care about data conflicts and ready to learn from your mistakes.
- Don't replace reports and operational systems
- When designing data warehouse, conform that the right tool is being used and don't spend much time on cleaning extracting and loading data.
- Identify and implement a training plan for the end users
- Try to involve all stakeholder and business partners in data warehouse implantation process.

Future of Data Warehousing

- The scope of data warehousing are on the rise as the regulatory constraints changes from time to time.
- It may limit the performance to combine different data from different sources. This may include unstructured data which is usually a difficult one to store.
- This is because the size of the data grows everyday data warehouse system helps the user to build and run data as a size grows

Conclusion

- A data warehouse is a database, which is kept separate from the organization's operational database.
- There is no frequent updating done in a data warehouse.
- It possesses consolidated historical data, which helps the organization to analyze its business.
- A data warehouse helps executives to organize, understand, and use their data to take strategic decisions.
- Data warehouse systems help in the integration of diversity of application systems.
- A data warehouse system helps in consolidated historical data analysis.
- Data mining functions such as association, clustering, classification, prediction can be integrated with OLAP operations to enhance the interactive mining of knowledge at multiple level of abstraction.

Thank You